Amplitude Modulation and Demodulation

Physical Measuring Techniques FK7063

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Introduction

Amplitude modulation (AM) is an important technique in communications systems, enabling the use of a specific carrier frequency to transmit signals of different frequencies.

In this experiment, an amplitude modulation circuit will be used to modulate a carrier voltage with a modulation signal and the resulting signal will be analyzed. The signal will be demodulated using a diode rectifier.

1 Theory

Amplitude modulation involves a carrier voltage $V_c(t) = A\cos\omega_c t$, and a modulation signal $S(t) = m\cos\omega_m t$, where the former must be of a higher frequency than the latter. An amplitude-modulated signal is obtained by adding the carrier multiplied by the modulation to the original carrier signal:

$$V(t) = V_c(t)(1 + S(t)) = A\sin\omega_c t(1 + m\cos\omega_m t) \tag{1}$$

We must have $m \leq 1$ so that the sign voltage is never flipped by the modulation. Using trigonometric identities, (1) can be brought to the form [1]

$$V(t) = A\cos\omega_c t + \frac{m}{2}A\left[\cos((\omega_c - \omega_m)t) + \cos((\omega_c + \omega_m)t)\right].$$

From this we can see that the fourier transform of the modulated signal will contain the original carrier frequency along with side bands shifted left and right by the amount of the modulation frequency. These side bands will never have more than half the amplitude of the carrier band since $m \leq 1$.

2 Setup

The main part of the amplitude modulation circuit is shown in fig. 1. The central com-

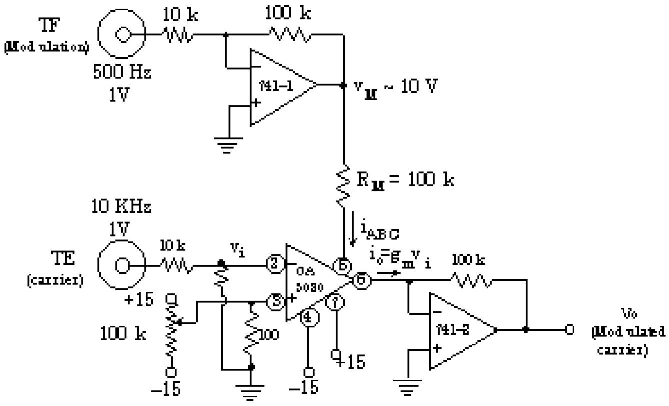

Figure 1: Schematic of the main part of the amplitude modulation circuit [1].

ponent is a CA3080 transconductance amplifier, which produces an output current proportional to the voltage v_i at its input and the Amplifier Bias Control current i_{ABC}. v_i comes from the carrier voltage TE and i_{ABC} is produced by amplifying the modulation voltage TF using a 741 operational amplifier (op amp; 741-1) and converting it to a current using a resistor. The output currect i_o is then proportional to the product of TE and TF and is converted to the desired modulated voltage using another op amp (741-2).

actually $V_c(1+V_m)$?

3 Conclusion

References

[1] *Amplitude Modulation and Demodulation*. Laboratory exercise manual. 2026.